Omar MOUCHTAKI

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Employment

New York University, Stern School of Business, New York, NY — 2024-present
Assistant Professor, Technology, Operations & Statistics

Education

Columbia University, Graduate School of Business, New York, NY — 2019-2024
Ph.D. in Operations Research, Decision, Risk and Operations division
Advisors: Prof. Omar Besbes and Prof. Will Ma

École Polytechnique, Palaiseau, France — 2016-2019
Bachelor of Science and Master of Science (Diplôme d'ingénieur) in Applied Mathematics

Lycée Louis-Le-Grand, Paris, France — 2014-2016

Journal Publications

1. "From Contextual Data to Newsvendor Decisions: On the Actual Performance of Data-Driven Algorithms"
with Omar Besbes and Will Ma. *Management Science*, Articles in Advance.

2. "Joint Assortment and Inventory Planning Under the Markov Chain Choice Model"
with Omar El Housni, Guillermo Gallego, Vineet Goyal, Salal Humair, Sangjo Kim, Ali Sadighian, Jingchen Wu. *Management Science*, vol. 72, no. 10, pp. 8667–8682, 2026.

3. "Beyond IID: Data-Driven Decision Making in Heterogeneous Environments"
with Omar Besbes and Will Ma. *Management Science*, vol. 71, no. 12, pp. 10538–10555, 2025.

4. "How Big Should Your Data Really Be? Data-Driven Newsvendor: Learning One Sample at a Time"
with Omar Besbes. *Management Science*, vol. 69, no. 10, pp. 5848-5865, 2023.
   - ★ `Finalist, Best OM Paper in Management Science, 2024`
   - ★ `First Place, RMP Jeff McGill Student Paper Award, 2021`
   - ★ `Finalist, INFORMS George Nicholson Student Paper Competition, 2021`
   - ★ `Finalist, APS Best Student Paper Award, 2021`

Working Papers

5. "Fast Revenue Maximization"
with Achraf Bahamou and Omar Besbes. Major revision in *M&SOM*.

6. "Prior-Independent Bidding Strategies for First-Price Auctions"
with Rachitesh Kumar. Major revision in *Operations Research*.

7. "Randomly Wrong Signals: Bayesian Auction Design with ML Predictions"
with Ilan Lobel and Humberto Moreira. Major revision in *Management Science*.

8. "What is the Value of Censored Data? An Exact Analysis for the Data-driven Newsvendor"
with Rachitesh Kumar. Working paper.

Refereed Conference Proceedings

1. "Prior-Independent Bidding Strategies for First-Price Auctions"
with Rachitesh Kumar. ACM Conference on Economics and Computation (EC), 2025.

2. "Beyond IID: Data-Driven Decision-Making in Heterogeneous Environments"
with Omar Besbes and Will Ma. Advances in Neural Information Processing Systems (NeurIPS), 2022.

Technical Reports

"Adversarial Feature Based Dynamic Pricing"
joint with Vineet Goyal. Technical Report, École Polytechnique (2019)
★ `École Polytechnique Research Prize in Computer Science, 2019`

Honors and Awards

Management Science Meritorious Service Award, 2025
Selected PhD Speaker, Columbia Business School EMBA & PhD Recognition Ceremony, 2025
Finalist, Best OM Paper in Management Science, 2024
Alex Woo MBA 1984 Doctoral Fellowship, 2023
Deming Center Doctoral Fellowship, 2022-2023 Doctoral fellow
Paul and Sandra Montrone Doctoral Fellowship, 2022
First Place, RMP Jeff McGill Student Paper Award, 2021
Finalist, INFORMS George Nicholson Student Paper Competition, 2021
Finalist, APS Best Student Paper Award, 2021
École Polytechnique Research Prize in Computer Science, 2019
Bronze Medalist, ACM International Collegiate Programming Contest, Southwestern Europe Regional Contest, 2017
Bronze Medalist, the 24th International Math Competition, 2017
French Government's Major-Excellence Scholarship, 2014

Grant

"Leveraging AI to Improve the Performance of the Distribution Network", Afriquia SMDC (Morocco). 2026-2027
NYU Stern Center for Sustainable Business Research Grant (co-PI: Jackie Baek), 2025

Industry Experience

Amazon.com, Applied Scientist Intern, New York, NY, USA — March 2021- May 2021
IBM Extreme Blue Program, Paris, France — Summer 2018

Teaching Experience

NYU Stern School of Business

Operations Management (Full-time MBA core)	Spring 2026
Operations Management (Part-time MBA core)	Spring 2025/ Spring 2026
Operations Management (Tech MBA core)	Fall 2025
Convex Optimization (PhD Core)	Fall 2026
Foundations of AI Agents (MBA Elective, course co-designer)	

Columbia Business School, **Teaching Assistant**

Operations Management (EMBA core)	Spring 2022
Business Analytics I (MBA core)	Fall 2021/Fall 2022/Spring 2024
Business Analytics II (MBA elective)	Summer 2020/Summer 2021

Professional Service

External

Journal reviewer: *Management Science*, *Operations Research*, *Manufacturing & Service Operations Management*, *Mathematics of Operations Research*
ACM Conference on Economics and Computation (EC) Program Committee Member (2024, 2025, 2026)
MSOM Conference SIG judge (2023, 2024, 2025, 2026)
MSOM Student Paper Competition judge (2025, 2026)
George Nicholson Student Paper Competition committee member (2026)

NYU Stern School of Business

Organizer, Stern OM Seminar Series, 2025
OM Recruiting Committee Member, 2024-2025
OM PhD admission committee, 2024-2026

Columbia Business School

Co-organizer of the NYC Operations Day PhD Colloquium, 2023
Brown Bag (DRO Internal Seminar) series co-organizer, 2021-2023
Columbia DRO PhD review Committee member, 2021

Other Activities

Board of Directors, École Polytechnique — 2017-2019
Elected student representative for the 2016 Class on the Board of Directors of Ecole Polytechnique.

Math&Maroc, Casablanca, Morocco — 2016-2019
Co-founder and vice-president of Math&Maroc, now a major Moroccan education NGO dedicated to promoting excellence in mathematics across Morocco.

Invited Talks

Duke, Fuqua School of Business, October 2026
UM6P, Moroccan Center for Game Theory, Rabat, Morocco, January 2026
Cornell University, ORIE School, January 2024
Imperial College, Business School, London, UK January 2024
NYU, Stern School of Business, January 2024
UT Dallas, Jindal School of Management, January 2024
IESE Business School, Barcelona, Spain, January 2024
IRIS conference, UM6P, Ben Guerir, Morocco, December 2023
CMU, Tepper School of Business, December 2023
University of Southern California, Marschall School of Business, December 2023
Rice, Business School, December 2023
INSEAD, Fontainebleau, France, November 2023
UCL, Business School, London, UK, November 2023
NUS, Business School, Singapore, November 2023
Northwestern, Kellogg School of Management, November 2023
Deming Center Board Meeting, May 2023
Columbia Economics Interdisciplinary Market Design Seminar, March 2023
Rotman OM&S Data Science Seminar, December 2022

Selected Presentations

What is the Value of Censored Data? An Exact Analysis for the Data-driven Newsvendor
MSOM Annual Conference, Boston — July 2026
RMP Annual Conference, Ann Arbor — July 2026

Auction Design using Value Prediction with Hallucinations
INFORMS Annual Meeting, Atlanta — Oct 2025
APS Conference, Atlanta — June 2025
Marketplace Innovation Workshop, Online — May 2025

Fast Revenue Maximization
RMP Annual Conference, New York — July 2025
INFORMS Annual Meeting, Seattle — Oct 2024

From Contextual Data to Newsvendor Decisions: On the Actual Performance of Data-Driven Algorithms
International Conference on Stochastic Programming, Paris, France — July 2025
INFORMS Annual Meeting, Phoenix — Oct 2023
Cornell Young Researchers Workshop, Ithaca — Oct 2023
RMP Annual Conference, London, UK — July 2023
MSOM Annual Conference, Montreal — June 2023
POMS Annual Conference, Orlando — May 2023
Deming Center Doctoral Fellow Seminar, New York — April 2023

Beyond IID: Data-Driven Decision-Making in Heterogeneous Environments
APS Conference, Nancy — June 2023
NeurIPS (Poster), New Orleans — December 2022
INFORMS Annual Meeting, Indianapolis — October 2022
CFOL Workshop at ICML (Poster), Baltimore — July 2022
RMP Annual Conference, Online — June 2022
Marketplace Innovation Workshop, Online — May 2022

How Big Should Your Data Really Be? Data-Driven Newsvendor: Learning One Sample at a Time
MSOM Supply Chain Management SIG Meeting, Munich, Germany — June 2022
INFORMS Annual Meeting, Anaheim — October 2021
MSOM Annual Conference, Online — June 2021
RMP Annual Conference, Online — June 2021
Amazon SCOT seminar, Online — May 2021

Joint Assortment and Inventory Planning under the Markov Chain Choice Model
MSOM Annual Conference, Online — June 2021
INFORMS Annual Meeting, Online — Nov 2020

Languages

English/French/Arabic: Fluent **Spanish:** Intermediate

Last modified: October 2026